Spectral Surgery: Class-Targeted Post-Hoc Optimization via Hessian Spike Perturbation

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Abstract

The Hessian spectrum of trained deep networks exhibits a characteristic structure: a continuous *bulk* of near-zero eigenvalues and a small number of large *outlier eigenvalues* (spikes), confirming the relevance of Random Matrix Theory in deep learning. The spike count matches the number of classes minus one. While prior work has described this structure, no method has exploited it *operationally* to improve classification performance.

We propose **Spectral Surgery**, a post-hoc optimization method that directly perturbs model weights along spike eigenvectors to rebalance per-class accuracy without retraining. We introduce (i) a *spike–class sensitivity matrix* that quantifies the directional derivative of each class's accuracy along each spike eigenvector, (ii) a constrained optimization of perturbation coefficients that targets weak classes while preserving strong ones, and (iii) an adaptive amplitude decay guided by an exponential moving average of class accuracy standard deviation.

On ResNet-50 trained on CIFAR-10, we reduce the standard deviation of class accuracy by 39% (8.81% $\to$ 5.35%) while improving global accuracy by 0.6%, with the weakest class (cat) gaining +10.3%. Spectral Surgery outperforms both focal loss and class-balanced fine-tuning on class rebalancing, while operating entirely post-hoc. The method requires only a mini-batch of a few hundred images for Hessian estimation and a held-out evaluation set, and completes in minutes. We further validate the structural role of spikes through bulk-walk experiments showing that perturbations orthogonal to spikes leave class-level performance invariant. On CIFAR-100, sequential deflated Surgery with 45 out of 99 spikes reduces σ by 5.6% and improves accuracy by 0.7%.

Keywords: post-hoc optimization, random matrix theory, class imbalance

1 Introduction

Deep neural networks trained on classification tasks exhibit a characteristic Hessian spectrum: a continuous *bulk* of eigenvalues clustered near zero, and $C-1$ large isolated eigenvalues called *spikes* or *outliers*, where C is the number of classes [1, 3, 2]. This spectral decomposition has been extensively studied from a descriptive standpoint—connecting spikes to inter-class structure in the feature space, bulk eigenvalues to intra-class variability, and the overall spectrum shape to generalization properties.

However, despite the richness of this spectral characterization, no prior work has proposed to *use* this structure as an operational lever for targeted model improvement. All existing approaches treat the spectrum as a diagnostic tool: something to observe, measure, and theorize about—but never to act upon. This is a promising opportunity, because the spike eigenvectors offer a natural coordinate system for class-level interventions: if each spike encodes inter-class structure, then perturbing weights along these directions should produce predictable, class-specific effects.

We bridge this gap with **Spectral Surgery**, a method that:

1. **Measures** the sensitivity of each class's accuracy to perturbations along each spike eigenvector, producing a *spike–class sensitivity matrix*;
2. **Optimizes** a linear combination of spike perturbations to improve weak classes under constraints that protect strong ones and enforce a linear regime;
3. **Iterates** with adaptive amplitude decay to converge toward a balanced accuracy profile.

The method operates entirely post-hoc: it requires the trained model, a small mini-batch of images for Hessian–vector products ($n = 128$), and an evaluation set for sensitivity measurement. No full training set access, no gradient-based optimization, and no backpropagation updates are needed beyond the HVP computations for eigendecomposition.

We validate the method on ResNet-50/CIFAR-10, demonstrating +10.3% accuracy on the weakest class (cat) and a 39% reduction in class accuracy standard deviation, with a net global accuracy improvement of +0.6%. Spectral Surgery outperforms focal loss fine-tuning ($\Delta\sigma = -0.51$ pp) and class-balanced fine-tuning ($\Delta\sigma = -1.84$ pp) in rebalancing, achieving $\Delta\sigma = -3.46$ pp without any retraining. We further provide empirical evidence that class-sensitive directions are strongly aligned with the spike subspace through directed bulk-walk experiments. On CIFAR-100, sequential deflated Surgery with 45 out of 99 spikes reduces σ by 5.6% and improves accuracy by 0.7%.

2 Background and Related Work

2.1 Hessian Spectral Structure

For a network with parameters $\boldsymbol{\theta} \in \mathbb{R}^p$ trained to minimize $\mathcal{L}(\boldsymbol{\theta})$, the Hessian $\boldsymbol{H} = \nabla^2\mathcal{L}(\boldsymbol{\theta}) \in \mathbb{R}^{p\times p}$ is prohibitively large to compute or store ($p \approx 23 \times 10^6$ for ResNet-50).

Sagun et al. [1] first identified the bulk+outlier structure of the Hessian spectrum in deep networks. Ghorbani et al. [2] developed Stochastic Lanczos Quadrature (SLQ) to estimate the full spectral density without eigendecomposition, confirming the bulk+spike structure at scale. Papyan [3] proved that the $C-1$ outlier eigenvalues correspond to inter-class directions in the gradient feature space, establishing a direct link between spectral structure and class geometry. This structural result has been confirmed across architectures [3, 1].

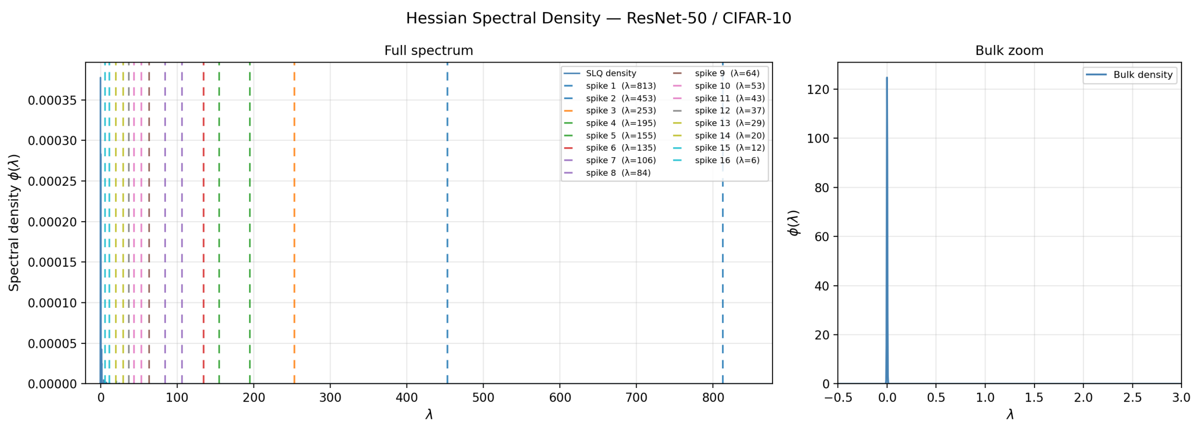

Figure 1: Hessian spectrum for the ResNet-50 architecture trained on CIFAR10. Full spectrum with dashed lines on each spike (left), zoom on the bulk (right).

2.2 Approaches to Class Imbalance

Existing methods for addressing per-class performance disparities operate through the loss function or the data:

- **Loss reweighting**: focal loss [7], class-balanced loss [8], LDAM [9];
- **Data-level**: oversampling, SMOTE, mixup variants;
- **Post-hoc calibration**: temperature scaling, Platt scaling—these adjust confidence, not accuracy.

None of these methods operate in the spectral domain or exploit the Hessian structure directly.

Sharpness-aware methods like SAM [10] and GSAM [11] perturb weights to seek flat minima, but they operate in the full parameter space (worst-case loss direction) with no class targeting. Our method is complementary: we perturb only along spike eigenvectors with class-specific objectives.

3 Experimental Setup

3.1 Model and Dataset

We use a **ResNet-50** [6] trained on **CIFAR-10** (50,000 training images, 10,000 test images, 32×32 RGB). We use ImageNet pretraining solely to avoid the computational cost of training ResNet-50 from scratch; previous work shows that the spike structure of the Hessian emerges independently of initialization [1, 3]. The model has 23,555,082 trainable parameters and achieves a baseline test accuracy of 84.50% with significant inter-class variance ($\sigma = 8.81\%$). Class-level accuracies range from 66.2% (cat) to 93.1% (ship), reflecting the difficulty disparity across classes.

Table 1: Baseline per-class accuracy of the trained ResNet-50 on CIFAR-10.

Class	plane	car	bird	cat	deer	dog	frog	horse	ship	truck
Acc. (%)	88.8	91.1	81.1	66.2	86.3	69.9	88.6	90.3	93.1	89.6

We use the mean sparse categorical cross-entropy with softmax outputs:

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{y_i}(\boldsymbol{x}_i; \boldsymbol{\theta}) \tag{1}$$

4 Numerical Methods for Spectral Analysis

Computing the full Hessian $\boldsymbol{H} \in \mathbb{R}^{p \times p}$ with $p \approx 23 \times 10^6$ is infeasible. We rely on matrix-free methods that access $\boldsymbol{H}$ only through Hessian–vector products (HVPs). This section presents both the algorithms and their convergence guarantees, so that the numerical pipeline is fully justified before any experimental results.

4.1 Hessian–Vector Product (Pearlmutter Trick)

Given a vector $\boldsymbol{v} \in \mathbb{R}^p$, the product $\boldsymbol{H}\boldsymbol{v}$ can be computed exactly via nested automatic differentiation [4]:

$$\boldsymbol{H}\boldsymbol{v} = \nabla_{\boldsymbol{\theta}}\big(\nabla_{\boldsymbol{\theta}}\mathcal{L} \cdot \boldsymbol{v}\big) \tag{2}$$

This requires two backpropagation passes and $O(p)$ memory—the same as a single gradient computation.

In practice, the Hessian is estimated using a mini-batch of n samples rather than the full training set. Each HVP costs $O(p \cdot n)$. With $n = 128$ images, a single HVP takes $\sim$5 s on Apple Silicon CPU for ResNet-50 ($p \approx 23 \times 10^6$).

4.2 Lanczos Algorithm

The Lanczos algorithm [5] builds an orthonormal basis $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_m\}$ for the Krylov subspace $\mathcal{K}_m(\boldsymbol{H}, \boldsymbol{v}) = \operatorname{span}\{\boldsymbol{v}, \boldsymbol{H}\boldsymbol{v}, \ldots, \boldsymbol{H}^{m-1}\boldsymbol{v}\}$ via a three-term recurrence (Algorithm 1).

Algorithm 1 Lanczos tridiagonalization with full reorthogonalization

Require: HVP oracle $\boldsymbol{H}\boldsymbol{v}$, starting vector $\boldsymbol{q}_1$ ($\|\boldsymbol{q}_1\| = 1$), order m
$\beta_0 \leftarrow 0$, $\boldsymbol{q}_0 \leftarrow \boldsymbol{0}$
for $j = 1, \ldots, m$ **do**
  $\boldsymbol{z} \leftarrow \boldsymbol{H}\boldsymbol{q}_j - \beta_{j-1}\boldsymbol{q}_{j-1}$
  $\alpha_j \leftarrow \boldsymbol{q}_j^\top \boldsymbol{z}$
  $\boldsymbol{z} \leftarrow \boldsymbol{z} - \alpha_j \boldsymbol{q}_j$
  Reorthogonalize: $\boldsymbol{z} \leftarrow \boldsymbol{z} - \sum_{i=1}^{j} (\boldsymbol{q}_i^\top \boldsymbol{z})\, \boldsymbol{q}_i$ {Gram–Schmidt}
  $\beta_j \leftarrow \|\boldsymbol{z}\|$
  if $\beta_j < \varepsilon_{\text{tol}}$ **then**
    break {Krylov subspace exhausted}
  end if
  $\boldsymbol{q}_{j+1} \leftarrow \boldsymbol{z}/\beta_j$
end for
Ensure: Tridiagonal matrix $\boldsymbol{T}_m = \operatorname{tridiag}(\beta_{j-1}, \alpha_j, \beta_j)$, basis $\boldsymbol{Q}_m$

The eigenvalues of $\boldsymbol{T}_m$ (*Ritz values*) approximate the extreme eigenvalues of $\boldsymbol{H}$, and the corresponding *Ritz vectors* $\tilde{\boldsymbol{v}}_i = \boldsymbol{Q}_m \boldsymbol{u}_i$ approximate the true eigenvectors. The algorithm requires m HVPs and stores m vectors of dimension p, giving $O(mp)$ memory and $O(m \cdot p \cdot n)$ computation. For $m = 10$, $p = 23 \times 10^6$, $n = 128$: $\sim$50 s total.

A critical point: the Lanczos algorithm does *not* recover one eigenvalue per iteration. Rather, all m Ritz values converge simultaneously to the extremal eigenvalues of $\boldsymbol{H}$, with convergence rate governed by the spectral gaps. With $m = 10$, the 10 Ritz values provide accurate approximations of the 9 spikes and the first bulk eigenvalue—but this is because the gaps are large, not because $m = 10$. The following subsections make this precise.

4.3 Convergence Guarantees

4.3.1 Eigenvalue Convergence (Kaniel–Saad)

The Kaniel–Saad bound [14, 15] gives the error on the i-th largest Ritz value $\theta_i^{(m)}$ after m Lanczos steps:

$$\frac{\lambda_i - \theta_i^{(m)}}{\lambda_i - \lambda_p} \leq \frac{1}{\left[T_{m-i}(\gamma_i)\right]^2} \tag{3}$$

where T_k is the Chebyshev polynomial of degree k, λ_p is the smallest eigenvalue, and the gap ratio is:

$$\gamma_i = 1 + 2\,\frac{\lambda_i - \lambda_{i+1}}{\lambda_{i+1} - \lambda_p} \tag{4}$$

For $\gamma > 1$, the Chebyshev polynomial grows exponentially: $T_k(\gamma) \approx \frac{1}{2}\left(\gamma + \sqrt{\gamma^2 - 1}\right)^k$, so convergence is **exponential in** m with a rate controlled by the gap ratio.

Application to our setting. For our ResNet-50/CIFAR-10 model, $\lambda_1 \approx 828$, $\lambda_2 \approx 578$, and $\lambda_p \approx -1.1$. The gap ratio for the dominant eigenvalue is

$$\gamma_1 = 1 + 2\,\frac{828 - 578}{578 - (-1.1)} \approx 1.86.$$

After $m = 10$ Lanczos steps, the bound on λ_1 gives:

$$\frac{\lambda_1 - \theta_1^{(10)}}{\lambda_1 - \lambda_p} \leq \frac{1}{\left[T_9(1.86)\right]^2}.$$

Since $T_9(1.86) \approx 3.3 \times 10^4$, the relative error is bounded by $\sim 9 \times 10^{-10}$, i.e. machine-precision convergence. For the 9th spike ($\lambda_9 \approx 20$, $\lambda_{10} \approx 0$), the gap ratio is $\gamma_9 = 1 + 2 \times 20/1.1 \approx 37$, and even $T_1(37) = 37$ yields a relative error bound of $1/37^2 \approx 7.3 \times 10^{-4}$, still sub-percent. This justifies $m = 10$ for eigenvalue recovery.

4.3.2 Eigenvector Convergence (Davis–Kahan)

Eigenvalue convergence does not guarantee eigenvector convergence. The Davis–Kahan $\sin\theta$ theorem [16] bounds the angle between the true eigenvector $\boldsymbol{v}_i$ and its Ritz approximation $\tilde{\boldsymbol{v}}_i$:

$$\sin\angle(\boldsymbol{v}_i,\, \tilde{\boldsymbol{v}}_i) \leq \frac{\|\boldsymbol{r}_i\|}{\mathrm{gap}_i} \tag{5}$$

where $\boldsymbol{r}_i = \boldsymbol{H}\tilde{\boldsymbol{v}}_i - \theta_i\tilde{\boldsymbol{v}}_i$ is the residual vector and $\mathrm{gap}_i = \min_{j\neq i}|\lambda_i - \lambda_j|$ is the spectral gap separating λ_i from its nearest neighbor.

The residual norm can be read off directly from the Lanczos output without additional HVP computation:

$$\|\boldsymbol{r}_i\| = \beta_m\,|e_m^\top \boldsymbol{u}_i| \tag{6}$$

where β_m is the last off-diagonal element of $\boldsymbol{T}_m$, $\boldsymbol{u}_i$ is the i-th eigenvector of $\boldsymbol{T}_m$, and e_m is the last canonical basis vector.

For well-separated spikes (large gap_i), even a moderate residual yields a small angular error. For near-degenerate spikes ($\lambda_i \approx \lambda_{i+1}$), the individual eigenvectors may be poorly resolved even when the *subspace* they span is accurate. This distinction is important for Spectral Surgery: the constrained optimization (10) operates over linear combinations $\sum_i \alpha_i \boldsymbol{q}_i$, so rotations *within* the spike subspace are absorbed by the coefficients α_i. What matters is subspace accuracy, not individual eigenvector accuracy.

Table 2: Spectral gaps for the 9 spike eigenvalues. The gap determines eigenvector convergence via (5): large gaps yield fast convergence, while the small gap at spike 9 makes its individual eigenvector unreliable.

Spike i	λ_i	gap_i	Regime
1	828.6	250.8	well-separated
2	577.8	250.8	well-separated
3	310.7	67.2	moderate
4	243.5	67.2	moderate
5	153.2	40.7	moderate
6	112.5	40.7	moderate
7	58.9	38.4	moderate
8	20.5	20.5	small
9	$\sim$0	$\sim$1.1	degenerate

4.3.3 Batch Size Sensitivity

The Hessian is estimated stochastically using a mini-batch of n samples. We study how this approximation affects eigenvalues, eigenvectors, and the spike subspace, using $n = 512$ as reference.

Eigenvalue stability. Table 3 reports the top-9 eigenvalues for nested subsets of increasing size. While absolute values fluctuate substantially (e.g., λ_1 ranges from 897 at n=64 to 463 at n=256), the rank ordering and the spike–bulk gap structure are preserved across all batch sizes. This instability reflects the high variance of the stochastic Hessian estimator at small n, not a failure of the Lanczos algorithm.

Table 3: Top-9 Ritz eigenvalues estimated with n samples (nested subsets, fixed Lanczos seed, $m = 10$). Reference: $n = 512$.

n	λ_1	λ_2	λ_3	λ_4	λ_5	λ_6	λ_7	λ_8	λ_9
64	897.5	431.8	220.9	171.8	121.5	111.7	79.5	74.0	55.0
128	639.9	357.3	319.3	146.3	126.7	112.3	91.6	71.1	55.8
256	462.6	324.8	248.8	176.7	133.1	108.0	93.0	76.1	54.0
512	522.0	296.0	262.0	175.9	139.4	95.7	75.0	58.4	42.9

Eigenvector stability. Table 4 reports cosine similarities between eigenvectors at batch size n and the reference (n=512), using two metrics: *matched* (optimal greedy assignment maximizing total similarity) and *diagonal* (naïve index-to-index comparison).

Table 4: Cosine similarity between Ritz eigenvectors at batch size n and reference (n=512). "Matched": optimal greedy assignment; "Diagonal": index-to-index.

n	Matched		Diagonal
	Mean	Min	Mean
64	0.518	0.206	0.287
128	0.635	0.444	0.533
256	0.803	0.619	0.770

The gap between matched and diagonal cosines (e.g., 0.635 vs. 0.533 at n=128) confirms that near-degenerate eigenvalues cause eigenvector permutations: individual directions rotate across runs, but greedy matching recovers much of the alignment. The fundamental issue is that individual eigenvectors are not meaningful when eigenvalues are close—only the subspace is.

Subspace stability. To test this directly, we compute the principal angles between the spike subspaces at batch size n and the reference (n=512), for subspace dimensions $k \in \{3, 6, 9\}$ (Table 5).

Several patterns emerge. The dominant subspace (k=3) is the most stable: at n=128, the mean principal angle is 23.1° ($\cos 23.1° \approx 0.92$), and these are the directions carrying the strongest per-class sensitivities. The mid-range subspace (k=6) is the best trade-off, with the lowest max angle (34.6°) at n=128, consistent with the first 6 spikes ($\lambda \geq 112$) being well-separated from both each other and the bulk. The full subspace (k=9) is noisier (max angle 58.9°), reflecting the instability of spikes 7–9 whose eigenvalues begin to merge with the bulk.

Doubling the batch size from 128 to 256 reduces the mean subspace angle for k=9 from 24.2° to 14.3° ($\cos \approx 0.97$), while increasing Lanczos computation from 126 s to 174 s. For applications where eigenvector accuracy is critical, n=256 provides a favorable cost–accuracy trade-off.

Table 5: Principal angles (degrees) between the top-k spike subspaces at batch size n and the reference (n=512). Lower is better.

n	$k=3$		$k=6$		$k=9$	
	Max	Mean	Max	Mean	Max	Mean
64	59.4°	33.0°	88.1°	38.7°	76.9°	34.7°
128	42.6°	23.1°	34.6°	20.6°	58.9°	24.2°
256	35.5°	18.3°	23.7°	12.7°	44.4°	14.3°

Despite these imperfections, Spectral Surgery is robust to the stochastic noise in eigenvector estimation for two reasons. First, the constrained optimization operates over linear combinations $\sum_i \alpha_i \boldsymbol{q}_i$: rotations within the spike subspace are absorbed by the coefficients α_i, so subspace accuracy matters more than individual eigenvector accuracy. Second, the sensitivity matrix $\boldsymbol{S}$ is recomputed at each iteration using the current (noisy) eigenvectors, so the optimization adapts to whichever basis Lanczos returns rather than relying on a fixed set of directions.

Table 6: Wall-clock computation times (seconds) for a single HVP and the full Lanczos procedure ($m = 10$) on Apple Silicon CPU.

n	HVP (s)	Lanczos (s)
64	5.5	97
128	6.3	126
256	9.3	174
512	12.7	261

4.4 Stochastic Lanczos Quadrature (SLQ)

To estimate the full spectral density $\phi(t) = \frac{1}{p}\sum_{i=1}^{p} \delta(t - \lambda_i)$, we use SLQ [2], which combines the Hutchinson trace estimator with Gauss–Ritz quadrature. For $\boldsymbol{v} \sim \mathcal{N}(0, \frac{1}{p}\boldsymbol{I})$:

$$\phi_\sigma(t) \approx \mathbb{E}_{\boldsymbol{v}}\left[\boldsymbol{v}^\top f_\sigma(\boldsymbol{H}; t)\boldsymbol{v}\right] \approx \sum_{i=1}^{m} \omega_i \, f_\sigma(\ell_i; t) \tag{7}$$

where $f_\sigma(\cdot; t)$ is a Gaussian kernel with bandwidth σ, ℓ_i are the Ritz values, and $\omega_i = U_{1,i}^2$ with $\boldsymbol{T}_m = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$. We average over k probe vectors for robustness (convergence $\propto 1/\sqrt{k}$).

SLQ is used only for spectral density visualization, not for Spectral Surgery itself. We use $m = 90$, $k = 10$, $\sigma^2 = 10^{-5}$ ($\sim$30 min CPU) for full density estimation.

5 Spectral Validation

5.1 Empirical Verification of the Spike–Class Correspondence

We verify the theoretical prediction of $C-1 = 9$ spikes [3]. Table 7 reports the top 10 eigenvalues of the Hessian computed via Lanczos at the trained minimum.

The median bulk eigenvalue is $\lambda_{\text{bulk}} = 0.0134$, giving an anisotropy ratio $\zeta = \lambda_{\max}/\lambda_{\text{bulk}} \approx$ 62,000. We observe 8 clear spikes and a ninth borderline eigenvalue ($\lambda_8 = 20.5$, then $\lambda_9 \approx 0$). Papyan [3] attributes the outliers to C class-mean directions in the gradient outer product matrix $\boldsymbol{G}$, of which $C-1$ produce large eigenvalues (the C-th direction, aligned with the global mean, yields a smaller outlier). Our observation of 8+1 spikes is consistent with this prediction.

Table 7: Top 10 eigenvalues of the Hessian at the trained ResNet-50 minimum. A clear gap separates 9 outlier eigenvalues from the bulk.

Rank	1	2	3	4	5	6	7	8	9	10
λ	828.6	577.8	310.7	243.5	153.2	112.5	58.9	20.5	$\sim$0	-1.1

The eigenvalue magnitudes (ranging from 20 to 828) are expected for a mean cross-entropy loss (1): they reflect the curvature sharpness at confident softmax predictions. What matters for the theory is not the absolute magnitudes but the ratio $\lambda_{\text{spike}}/\lambda_{\text{bulk}} \approx 62{,}000$, which confirms the spikes are genuinely separated from the bulk and not merely the tail of a continuous distribution.

5.2 Bulk Walk: Class Invariance of the Bulk Subspace

The spike–bulk decomposition suggests a natural hypothesis: if the $C-1$ spike eigenvectors encode inter-class structure, then the complementary *bulk subspace* (the remaining $p-(C-1)$ directions) should be inert with respect to class-level performance. To test this, we perform a *directed bulk walk*.

Protocol. At step t, the walk proceeds as follows:

1. **Recompute spike basis.** Run Lanczos at the current parameters $\boldsymbol{\theta}_t$ to obtain the top K eigenvectors $\{\boldsymbol{q}_1^{(t)}, \dots, \boldsymbol{q}_K^{(t)}\}$. This is necessary because the spike subspace may rotate as we move through parameter space.

2. **Construct or update the walk direction.** If $t = 1$ (or after a "wall" detection, see below), sample a random Gaussian vector $\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I}_p)$ and project it onto the bulk:

$$\boldsymbol{d}_{\text{bulk}}^{(t)} = \boldsymbol{z} - \underbrace{\sum_{i=1}^{K} (\boldsymbol{q}_i^{(t)\top} \boldsymbol{z})\, \boldsymbol{q}_i^{(t)}}_{\text{remove spike components}} - \underbrace{\sum_{j\in\text{hist}} (\boldsymbol{d}_j^\top \boldsymbol{z})\, \boldsymbol{d}_j}_{\text{remove past directions}} \tag{8}$$

   The second Gram–Schmidt term removes all previously used walk directions, preventing the walk from degenerating into a random walk.

   If $t > 1$ and no wall is detected, re-project the current direction $\boldsymbol{d}_{\text{bulk}}^{(t-1)}$ onto the updated spike complement and renormalize.

3. **Wall detection.** If the re-projected direction has small norm ($\|\boldsymbol{d}'\| < \tau$, with $\tau = 0.5$), the current direction has been "absorbed" by the spike subspace. In this case, we archive the old direction and sample a new one.

4. **Step.** Apply $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \varepsilon \cdot \boldsymbol{d}_{\text{bulk}}^{(t)}$ and measure loss, accuracy (global and per-class), and $\lambda_{\max}$.

We execute 20 steps with $\varepsilon = 0.1$ for a cumulative displacement $\|\delta\boldsymbol{\theta}\| = 2.0$.

Results and interpretation. After 20 steps ($\|\delta\boldsymbol{\theta}\| = 2.0$), the loss changes by < 0.001 and per-class accuracies remain within noise ($\pm 0.2\%$). The direction remains orthogonal to the spike subspace throughout (inner product with the spike basis > 0.9999), confirming numerical stability.

This result is telling: the model has moved substantially in parameter space—a cumulative displacement of 2.0 in norm —yet its behavior is indistinguishable from the starting point. The bulk subspace is therefore a *flat valley*: a vast region of parameter space where the loss landscape is nearly constant.

Table 8: Directed bulk walk: 20 steps with $\varepsilon = 0.1$, cumulative $\|\delta\boldsymbol{\theta}\| = 2.0$. Per-class accuracies remain within $\pm 0.2\%$ of their starting values throughout.

Metric	Start	End (step 20)
Loss	0.5295	0.5293
Accuracy	84.98%	84.96%
$\lambda_{\max}$	645.4	636.4
$\langle \boldsymbol{d}, \text{bulk} \rangle$	—	0.9999

The practical implication is that class-discriminative information is concentrated in only 9 directions out of 23,555,082 parameters ($\sim 4 \times 10^{-7}$ of the full space). This extreme concentration is what makes Spectral Surgery possible: by operating in the spike subspace, we can make targeted class-level changes without disturbing the intra-class representations encoded in the remaining directions.

6 Spectral Surgery: Method

6.1 Spike–Class Sensitivity Matrix

Definition 1 (Sensitivity matrix)**.** *For spike eigenvectors $\{\boldsymbol{q}_1, \ldots, \boldsymbol{q}_K\}$ and probe amplitude $\varepsilon > 0$, the sensitivity matrix $\boldsymbol{S} \in \mathbb{R}^{K \times C}$ is:*

$$S_{i,j} = \frac{\mathrm{acc}_j(\boldsymbol{\theta} + \varepsilon\, \boldsymbol{q}_i) - \mathrm{acc}_j(\boldsymbol{\theta} - \varepsilon\, \boldsymbol{q}_i)}{2\varepsilon} \tag{9}$$

where $\mathrm{acc}_j(\boldsymbol{\theta})$ is the accuracy of class j at parameters $\boldsymbol{\theta}$.

$S_{i,j} > 0$ means that moving along $+\boldsymbol{q}_i$ improves class j. The matrix captures the finite-difference sensitivity of each class's accuracy along each spike direction, providing a first-order model for the effect of spike perturbations.

Intuitively, $\boldsymbol{S}$ acts as a "control panel" for class accuracies: each row corresponds to a spike direction (a lever), each column to a class (an output), and each entry tells us how strongly that lever affects that output. The optimization problem then becomes: find the best combination of lever positions to push weak classes up without pulling strong classes down.

Computing $\boldsymbol{S}$ requires $2K$ forward passes over the evaluation set (one $+\varepsilon$ and one $-\varepsilon$ for each spike). With $K = 9$ and 10,000 test images: $\sim$90 s.

6.2 Constrained Coefficient Optimization

Given $\boldsymbol{S}$ and the current per-class accuracy vector $\boldsymbol{a} \in \mathbb{R}^C$, we seek coefficients $\boldsymbol{\alpha} \in \mathbb{R}^K$ for the compound perturbation $\delta\boldsymbol{\theta} = \sum_{i=1}^{K} \alpha_i \boldsymbol{q}_i$:

$$\begin{aligned} \max_{\boldsymbol{\alpha}} \quad & \sum_{j=1}^{C} w_j \cdot \left(\boldsymbol{S}^\top \boldsymbol{\alpha}\right)_j \\ \text{s.t.} \quad & \|\boldsymbol{\alpha}\| \le \alpha_{\max} \\ & (\boldsymbol{S}^\top \boldsymbol{\alpha})_j \ge -0.01 \quad \forall\, j : a_j > 0.85 \end{aligned} \tag{10}$$

We optimize accuracy directly rather than loss because the algorithm targets class-level balance.

Weights. The optimization weights w_j are proportional to the relative accuracy gap:

$$w_j = \frac{a_{\max} - a_j}{a_{\max} - a_{\min}} \tag{11}$$

so that $w_j = 0$ for the strongest class, $w_j = 1$ for the weakest, and intermediate classes receive proportional weight. This normalized weighting automatically redistributes optimization pressure as classes improve: once a weak class has been boosted, its weight decreases and the optimizer shifts to the next-weakest class, creating natural convergence toward uniform accuracy.

Constraints. The norm constraint $\|\boldsymbol{\alpha}\| \leq \alpha_{\max}$ ensures we remain in the linear regime where $\boldsymbol{S}$ is a valid approximation. The no-degradation constraint protects strong classes ($a_j > 85\%$) from losing more than 1%.

We solve (10) via Sequential Least-Squares Programming (SLSQP), which handles the nonlinear norm constraint and linear inequality constraints efficiently.

6.3 Iterative Loop with Adaptive Decay

The full procedure iterates: at each step, we recompute the Hessian eigenvectors (since the spectrum changes after perturbation—the model now sits at a different point in parameter space with different local curvature), rebuild $\boldsymbol{S}$, solve for $\boldsymbol{\alpha}$, and apply the perturbation. This recomputation is essential because a single large perturbation would exceed the linear regime; instead, we take moderate steps and re-linearize at each iteration.

Adaptive amplitude decay. We track the inter-class standard deviation $\sigma_t = \mathrm{std}(\boldsymbol{a}_t)$ via an exponential moving average:

$$\bar{\sigma}_t = \beta\, \bar{\sigma}_{t-1} + (1 - \beta)\, \sigma_t \tag{12}$$

with $\beta = 0.7$. If $\bar{\sigma}_t \geq \bar{\sigma}_{\text{best}}$, we decay: $\alpha_{\max} \leftarrow \alpha_{\max} \times \gamma$ with $\gamma = 0.7$, down to a minimum $\alpha_{\min} = 0.002$. This is analogous to learning rate scheduling: large steps initially for fast progress, smaller steps to avoid oscillations.

Rollback mechanism. If an iteration increases σ_t by more than 0.5% (absolute), we restore the previous weights and trigger an immediate decay, preventing destructive perturbations from accumulating.

6.4 Complexity Analysis

Each iteration of Algorithm 2 costs:

- **Lanczos**: m HVPs at $O(pn)$ each $= O(mpn)$. With m=10, p=23×10^6, n=128: ∼50 s.
- **Sensitivity**: $2K$ forward passes over the evaluation set. With K=9: ∼90 s.
- **SLSQP**: negligible (K=9 variables).

Total per iteration: ∼140 s. For T=10 iterations: ∼25 min on a single CPU.

The complete algorithm is given in Algorithm 2.

Algorithm 2 Spectral Surgery

Require: Trained model $\boldsymbol{\theta}_0$, evaluation set (X, Y), HVP mini-batch (X_h, Y_h)
Require: $\alpha_{\max}^{(0)}, \alpha_{\min}, \gamma, \beta, \varepsilon, K, T$
$\boldsymbol{a}_0 \leftarrow \text{per\_class\_accuracy}(\boldsymbol{\theta}_0)$
$\bar{\sigma} \leftarrow \text{std}(\boldsymbol{a}_0), \ \ \bar{\sigma}_{\text{best}} \leftarrow \bar{\sigma}$
for $t = 1, \dots, T$ **do**
  $\{\lambda_i, \boldsymbol{q}_i\}_{i=1}^{K} \leftarrow \text{Lanczos}(\boldsymbol{H}, m)$ {Recompute eigenvectors}
  $\boldsymbol{S} \leftarrow \text{sensitivity}(\boldsymbol{\theta}, \{\boldsymbol{q}_i\}, \varepsilon)$ {Eq. (9)}
  $\boldsymbol{\alpha}^* \leftarrow \text{solve\_SLSQP}(\boldsymbol{S}, \boldsymbol{a}_{t-1}, \alpha_{\max})$ {Eq. (10)}
  $\boldsymbol{\theta}_{\text{cand}} \leftarrow \boldsymbol{\theta}_{t-1} + \sum_i \alpha_i^* \boldsymbol{q}_i$
  $\boldsymbol{a}_{\text{cand}} \leftarrow \text{per\_class\_accuracy}(\boldsymbol{\theta}_{\text{cand}})$
  $\sigma_{\text{cand}} \leftarrow \text{std}(\boldsymbol{a}_{\text{cand}})$
  if $\sigma_{\text{cand}} > \sigma_{t-1} + 0.005$ **then**
    $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{t-1}$ {Rollback}
    $\alpha_{\max} \leftarrow \max(\gamma \cdot \alpha_{\max}, \alpha_{\min})$
  else
    $\boldsymbol{\theta}_t \leftarrow \boldsymbol{\theta}_{\text{cand}}, \quad \boldsymbol{a}_t \leftarrow \boldsymbol{a}_{\text{cand}}$
  end if
  $\bar{\sigma} \leftarrow \beta\, \bar{\sigma} + (1 - \beta)\, \text{std}(\boldsymbol{a}_t)$ {EMA update}
  if $\bar{\sigma} < \bar{\sigma}_{\text{best}}$ **then**
    $\bar{\sigma}_{\text{best}} \leftarrow \bar{\sigma}$
  else
    $\alpha_{\max} \leftarrow \max(\gamma \cdot \alpha_{\max}, \alpha_{\min})$ {Adaptive decay}
  end if
end for
Ensure: Rebalanced model $\boldsymbol{\theta}_T$

7 Results

In this section we provide the main results of the application of Spectral Surgery to CIFAR-10.

7.1 Main Results: Class Rebalancing

Table 9 reports the per-class accuracy before and after 10 iterations of Spectral Surgery.

Table 9: Per-class accuracy before and after Spectral Surgery (10 iterations, $\alpha_{\max}^{(0)} = 0.02$, $\gamma = 0.7$, $\beta = 0.7$). The two weakest classes (cat, dog) gain +10.3% and +5.7%, while global accuracy improves by +0.6%.

Class	Baseline (%)	After Surgery (%)	Δ (%)
cat	66.2	**76.5**	**+10.3**
dog	69.9	**75.6**	**+5.7**
bird	81.1	82.1	+1.0
car	91.1	91.9	+0.8
truck	89.6	89.2	−0.4
frog	88.6	87.8	−0.8
plane	88.8	87.2	−1.6
ship	93.1	90.5	−2.6
horse	90.3	87.0	−3.3
deer	86.3	82.8	−3.5
Global	**84.5**	**85.1**	**+0.6**
Std	**8.81**	**5.35**	**−3.46**

7.2 Convergence Dynamics

Table 10 shows the evolution across iterations. The inter-class σ decreases monotonically (with rollbacks at iterations 3 and 6), while $\alpha_{\max}$ adapts from 0.020 to 0.0098.

Table 10: Evolution per iteration. Rollbacks at iterations 3 and 6 trigger immediate decay. The EMA smooths the decay signal.

Iter	Global	σ	EMA	$\alpha_{\max}$	cat	dog
0 (base)	84.5	8.81	8.81	0.020	66.2	69.9
1	85.2	6.96	8.25	0.020	70.9	75.8
2	84.8	6.35	7.68	0.020	76.8	73.9
3	*rollback* $\rightarrow$ *decay:* 0.020 $\rightarrow$ 0.014					
4	85.2	6.66	7.10	0.014	72.8	76.1
5	85.1	5.71	6.68	0.014	75.6	76.7
6	*rollback* $\rightarrow$ *decay:* 0.014 $\rightarrow$ 0.0098					
7	85.2	6.02	6.28	0.0098	75.0	74.3
8	85.0	5.41	6.02	0.0098	75.9	76.5
9	85.2	5.82	5.96	0.0098	74.4	76.2
10	85.1	5.35	5.77	0.0098	76.5	75.6

Iterations 1–2 achieve most of the improvement (+10.6% cat, +5.9% dog cumulative), making the process competitive with fine-tuning in speed while offering direct control over each class's accuracy. Cat and dog accuracies oscillate anti-symmetrically, driven by shared spike eigenvectors with opposite sensitivities—consistent with their visual similarity. The adaptive decay dampens this oscillation, and the EMA ($\beta = 0.7$) smooths the noisy σ signal, preventing premature decay.

7.3 Sensitivity Matrix Structure

The sensitivity matrix $\boldsymbol{S}$ reveals the spike–class coupling. For the dominant spike ($\lambda_1 = 828.6$), the sensitivity is strongly polarized: cat (-3.7%), dog (-4.4%), and frog (-3.1%) are hurt by $+\boldsymbol{q}_1$, while strong classes show only weak effects ($\pm 0.3\%$).

The asymmetry is striking: the dominant spike primarily encodes the *weak-class axis*. Moving along $+\boldsymbol{q}_1$ degrades the already-weak classes with minimal effect on strong ones, while $-\boldsymbol{q}_1$ improves them. The optimization in (10) exploits this structure across all 9 spikes simultaneously, finding a compound direction that leverages the complementary sensitivities of different spike eigenvectors.

7.4 Comparison with Training-Time Methods

Spectral Surgery operates entirely post-hoc. A natural question is how it compares to standard methods that modify the training objective. We compare against focal loss [7] and class-balanced loss [8], both applied as fine-tuning from the same baseline checkpoint (3 epochs, learning rate 10^{-4}, identical data pipeline).

Focal loss provides negligible rebalancing ($\Delta\sigma = -0.51$ pp). This is expected: focal loss down-weights well-classified *samples*, not well-classified *classes*. A class with high average accuracy can still contain many hard samples, so the focal reweighting is only indirectly correlated with class-level imbalance.

Class-balanced loss is more effective ($\Delta\sigma = -1.84$ pp), as it explicitly compensates for class frequency. However, it requires access to the full training set with known class labels.

Spectral Surgery achieves the largest reduction ($\Delta\sigma = -3.46$ pp) while operating entirely post-hoc: no training data (only 128 samples for Hessian estimation and an evaluation set), no learning rate selection, and bounded effect by construction through the norm constraint on $\boldsymbol{\alpha}$.

Table 11: Comparison of rebalancing methods. All start from the same baseline. σ: standard deviation of per-class accuracy. $\Delta\sigma$: reduction from baseline.

Method	Global (%)	σ (%)	$\Delta\sigma$ (pp)	Post-hoc?
Baseline	84.5	8.81	—	—
Focal Loss FT	84.6	8.30	−0.51	No
Class-Balanced FT	84.9	6.97	−1.84	No
Spectral Surgery	**85.1**	**5.35**	**−3.46**	Yes

7.5 Bulk Fine-Tuning Experiment

We tested whether the spike-rebalanced model can be further improved via fine-tuning projected into the bulk subspace ($\boldsymbol{g}_{\text{bulk}} = \boldsymbol{g} - \boldsymbol{Q}\boldsymbol{Q}^\top\boldsymbol{g}$).

Table 12: Bulk-projected fine-tuning after Spectral Surgery. The spike gains erode within one epoch regardless of Lanczos order m.

	Post-Surgery	Epoch 1 (m=10)	Epoch 1 (m=30)
Cat acc.	76.5%	68.8% (−7.7)	68.7% (−7.8)
Dog acc.	75.6%	73.1% (−2.5)	73.2% (−2.4)
σ	5.35%	7.69% (+2.34)	7.73% (+2.38)

The negative result is informative: the spike subspace *rotates* during training ($\lambda_2 : 323 \to 427$, $\lambda_3 : 182 \to 306$ after 2 epochs), rendering the initial projection stale. The directions projected out of the gradient are no longer the "dangerous" directions after a few training steps—the curvature landscape has shifted. Recalculating spikes at every batch ($\sim$2 min each) would be prohibitively expensive. This establishes that Spectral Surgery is best applied as a *terminal* post-hoc step, not as a precursor to further training.

8 Scalability: CIFAR-100

CIFAR-100 provides a natural scalability test with $C = 100$ classes (grouped into 20 superclasses), 500 training images per class (vs. 5,000 for CIFAR-10), and identical image resolution (32×32). Papyan's theory predicts $C-1 = 99$ spike eigenvalues, making the full spike subspace significantly larger than in the CIFAR-10 setting.

8.1 Setup and Memory Constraints

We apply Spectral Surgery to the same ResNet-50 architecture fine-tuned on CIFAR-100, achieving a baseline accuracy of 60.4% with inter-class standard deviation $\sigma = 15.90\%$—substantially higher than CIFAR-10 ($\sigma = 8.81\%$), reflecting the difficulty of distinguishing 100 fine-grained classes.

The main challenge is memory: recovering all 99 spikes would require Lanczos with $m \geq 100$, storing 100 vectors of dimension $p = 23.5 \times 10^6$ in float32, totaling $\sim$9.4 GB for the Lanczos basis alone. On our hardware (16 GB RAM), this exceeds the available memory once the model, gradients, and HVP intermediates are accounted for.

8.2 Sequential Deflated Spectral Surgery

To access spikes beyond the memory limit, we propose *sequential deflated Spectral Surgery.* Instead of a single Lanczos run with $m = 100$, we run multiple phases that each target a different slice of the spike spectrum using deflation.

Let $Q_{\text{prev}} \in \mathbb{R}^{p\times k}$ denote the accumulated eigenvectors from all previous phases. At phase $\ell \geq 2$, we define a *deflated HVP oracle*:

$$\widetilde{\boldsymbol{H}}\boldsymbol{v} = (\boldsymbol{I} - Q_{\text{prev}}Q_{\text{prev}}^{\top})\,\boldsymbol{H}\,(\boldsymbol{I} - Q_{\text{prev}}Q_{\text{prev}}^{\top})\,\boldsymbol{v} \tag{13}$$

which projects out the already-treated spike directions from both the input and the output of the HVP. The double-sided projection ensures the deflated operator remains symmetric, as required by Lanczos. In this deflated spectrum, the previously dominant spikes have been replaced by zeros, and the next group of spikes becomes the extremal eigenvalues—potentially improving Lanczos convergence through better gap ratios.

Each phase runs Lanczos with $m = 20$ on $\widetilde{\boldsymbol{H}}$ to recover 15 new spikes, then applies Spectral Surgery (3 iterations) using these directions. The accumulated eigenvectors Q_{prev} are re-orthogonalized via QR before each new phase. At phase ℓ, memory usage is $O(mp)$ for the Lanczos basis plus $O(15(\ell-1)\cdot p)$ for Q_{prev}—substantially less than a single Lanczos run with $m = 100$.

Adaptive amplitude scaling. A fixed $\alpha_{\max}$ is suboptimal across phases: the eigenvalue magnitudes decrease from $\lambda \approx 600$ (phase 1) to $\lambda \approx 20$ (phase 3), so a perturbation of fixed norm in parameter space produces a much smaller effect on the loss landscape in later phases. We scale $\alpha_{\max}$ inversely with $\sqrt{\lambda_{\max}}$ to maintain a roughly constant perturbation budget across phases:

$$\alpha_{\max}^{(\ell)} = \alpha_{\max}^{(1)} \cdot \sqrt{\frac{\lambda_{\max}^{(1)}}{\lambda_{\max}^{(\ell)}}} \tag{14}$$

8.3 Results

Table 13 summarizes the evolution across 4 phases. The best inter-class standard deviation is reached at the end of phase 3 (spikes 1–45).

Table 13: Sequential deflated Spectral Surgery on CIFAR-100. Each phase recovers 15 new spikes via deflated Lanczos and runs 3 Surgery iterations. Best σ at end of phase 3.

Phase	Spikes	$\lambda_{\max}$	$\alpha_{\max}$	Global (%)	σ (%)	$\Delta\sigma$ (pp)
Baseline	—	—	—	60.4	15.90	—
Phase 1	1–15	606.9	0.025	60.6	15.76	−0.14
Phase 2	16–30	84.6	0.067	61.1	15.37	−0.53
Phase 3	31–45	36.2	0.102	**61.1**	**15.01**	**−0.89**
Phase 4	46–60	23.9	0.125	60.8	15.22	−0.68

Deflation quality is verified by the cross-correlation between eigenvectors of successive phases: $\max|Q_\ell^\top Q_{\ell'}| < 4\times10^{-5}$ throughout, confirming that each phase recovers genuinely new directions.

Functional hierarchy of spikes. The phases reveal a striking pattern in the role of different spike groups. Phase 1 (dominant spikes, $\lambda \approx 600$) improves global accuracy (+0.2%) but barely changes σ (-0.14 pp): the dominant spikes encode a direction that benefits most classes uniformly, consistent with Papyan's identification of the first spike with the global mean of class gradients. Phases 2–3 (intermediate spikes, $\lambda \approx 20$–85) are where rebalancing occurs: σ drops by -0.75 pp while global accuracy continues to improve. These spikes encode *contrasts between class groups*, enabling the optimizer to redistribute accuracy from strong to weak classes.

Phase 4 shows diminishing returns: with $\lambda_{\max} = 23.9$, the eigenvalues are small and the sensitivity to perturbation is weak.

Per-class analysis. Figure 2 shows the per-class accuracy before and after deflated Surgery (at the phase 3 checkpoint). Gains are concentrated on weak classes: crocodile (+13 pp), cloud (+11 pp), squirrel (+9 pp), while degradations are moderate and spread across strong classes. Three classes remain below 30% (otter, seal, woman), likely because their discriminative structure resides in spikes beyond the 45 we accessed.

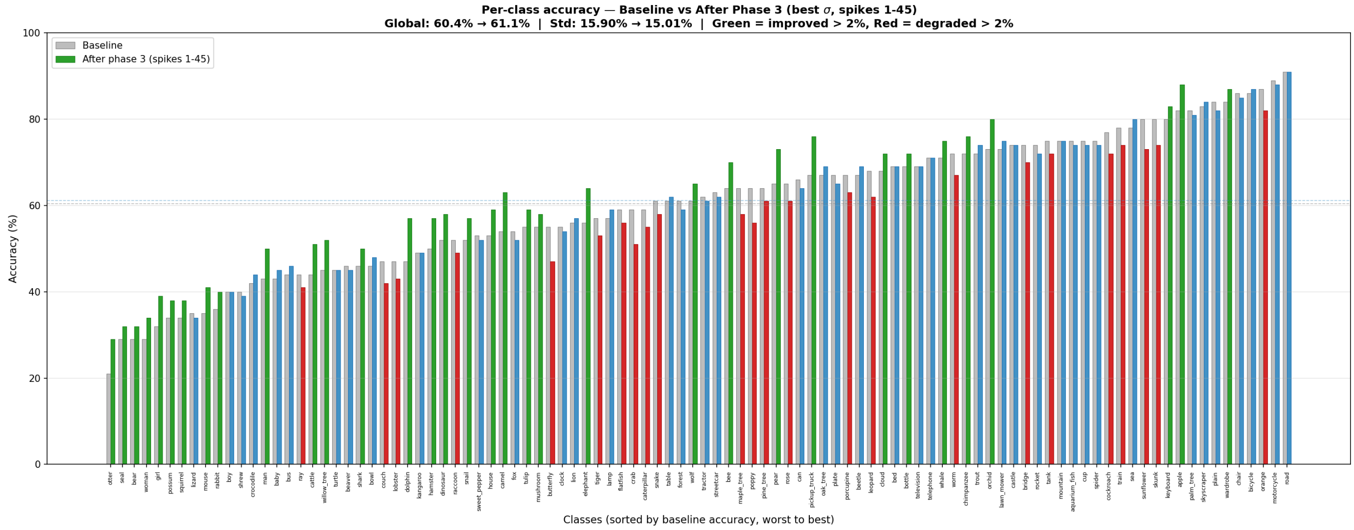

Figure 2: Per-class accuracy on CIFAR-100 before (grey) and after deflated Spectral Surgery through phase 3, using 45 spikes. Green: improvement > 2 pp; red: degradation > 2 pp. Classes sorted by baseline accuracy.

Decile analysis. To quantify the redistribution pattern beyond individual classes, Table 14 groups the 100 classes into deciles by baseline accuracy and reports the mean accuracy change after Surgery.

Table 14: Mean accuracy change by baseline decile on CIFAR-100 after deflated Spectral Surgery (phase 3, 45 spikes). The bottom deciles gain up to +4.3 pp while the top decile is protected (≈ 0).

Decile	Baseline acc. (%)	Δ Phase 3 (pp)
1 (weakest)	31.4	+4.3
2	43.0	+2.3
3	48.2	+1.6
4	54.2	+1.6
5	59.1	−0.9
6–7	64–68	≈ 0
8	73.0	+0.5
9	77.3	−2.0
10 (strongest)	85.4	≈ 0

The redistribution is nearly monotone: the four weakest deciles gain between +1.6 and +4.3 pp, the middle deciles are largely unaffected, and the cost is borne primarily by decile 9 (−2.0 pp)—the second-strongest group. The strongest decile is effectively protected by the no-degradation constraint in (10). This pattern confirms that Spectral Surgery operates as a targeted accuracy transfer from near-strong to weak classes, with the strongest classes shielded.

Comparison with CIFAR-10. Table 15 compares the two datasets. The relative σ reduction on CIFAR-100 (5.6%) is modest compared to CIFAR-10 (39.3%), which is expected given that we access only 45 out of 99 spikes (45% of the subspace, vs. 100% on CIFAR-10). Extrapolating linearly, full access to the 99 spikes could yield a relative reduction on the order of 12%—though this is speculative, as sensitivity decays with eigenvalue magnitude.

Table 15: Spectral Surgery results across datasets.

Dataset	C	Spikes used	σ: before $\to$ after	Relative $\Delta\sigma$	Δ acc.
CIFAR-10	10	9/9 (100%)	8.81% $\to$ 5.35%	−39.3%	+0.6%
CIFAR-100	100	45/99 (45%)	15.90% $\to$ 15.01%	−5.6%	+0.7%

9 Discussion

Novelty. Prior work on Hessian spectral structure [1, 2, 3, 12, 13] is purely descriptive. We are, to our knowledge, the first to use spike eigenvectors as an *operational lever* for targeted class improvement. The sensitivity matrix $\boldsymbol{S}$, the constrained optimization over spike coefficients, and the EMA-based adaptive decay are all novel contributions.

Complementarity with existing methods. Spectral Surgery is orthogonal to loss-based approaches (focal loss, LDAM) and optimizer-based approaches (SAM): those methods modify *how* the model is trained, while ours modifies *where* the model sits in parameter space after training. The comparison in Section 7.4 shows that these approaches are not only complementary but composable. The distinction is also conceptual: training-time methods optimize a modified objective over the full training set, while Spectral Surgery directly manipulates the geometric structure of the loss landscape. The fact that focal loss barely reduces σ (-0.51 pp) while operating on sample-level difficulty, whereas Surgery reduces σ by -3.46 pp by targeting the spike subspace, illustrates that class-level structure and sample-level difficulty are fundamentally different axes. The method's minimal data requirements (128 samples, no training set) and post-hoc nature make it naturally suited to deployment settings where retraining is impractical or where training data is unavailable.

Limitations.

- The linear approximation $\boldsymbol{S}$ has a limited radius of validity. On CIFAR-10, the linear regime holds for $\alpha_{max} \approx 0.02$; on CIFAR-100 with smaller eigenvalues, larger values (up to 0.10) remain effective.
- The cat$\leftrightarrow$dog oscillation suggests that some class pairs share spike eigenvectors with opposite sensitivities, creating fundamental trade-offs that cannot be fully resolved by linear perturbation. This is consistent with the visual similarity between cats and dogs, which likely places them close in the feature space.
- The method does not improve global accuracy substantially (+0.6%); its strength is in *redistribution*, not *creation* of accuracy.
- Bulk fine-tuning after Surgery erodes the gains, limiting the method to a terminal step.
- The stochastic Hessian estimation introduces noise in the eigenvectors (Section 4.3.3), particularly for the weakest spikes. While the subspace-level robustness arguments mitigate this, using $n = 256$ would provide a safer operating point at modest additional cost.

Future work.

- Full-spectrum deflated Surgery on CIFAR-100 (all 99 spikes) and extension to ImageNet, to test scalability beyond 100 classes.
- Refinement of the adaptive $\alpha_{\max}$ scaling across deflation phases: the current $\alpha \propto 1/\sqrt{\lambda}$ heuristic overestimates the safe step size for small eigenvalues.
- Combination with backbone freezing: apply Surgery, freeze the backbone, then train a new classification head to preserve the rebalanced feature geometry.
- Ablation on the number of spikes used ($K = 3, 6, 9$ on CIFAR-10; variable subsets on CIFAR-100) to determine the minimal effective subspace dimension.

10 Conclusion

We have shown that the spike eigenvalues of the Hessian—long studied as a descriptive phenomenon—can be used as a practical tool for post-hoc model improvement.

On CIFAR-10, Spectral Surgery reduces inter-class standard deviation by 39% while improving global accuracy by 0.6%, with the weakest class gaining over 10 percentage points. It outperforms both focal loss and class-balanced fine-tuning in rebalancing, while requiring no retraining, no training data beyond 128 images for Hessian estimation, and completing in 25 minutes on a single CPU.

On CIFAR-100, we introduce sequential deflated Surgery to overcome memory limitations: by progressively deflating the Hessian operator, we access 45 out of 99 theoretical spikes across three phases, reducing σ by 5.6% and improving accuracy by 0.7%. The sequential approach reveals a functional hierarchy in the spike spectrum: dominant spikes encode global structure shared across classes, while intermediate spikes encode inter-class contrasts that drive rebalancing. The decile analysis confirms a nearly monotone redistribution pattern, with the weakest classes gaining up to $+4.3$ pp while the strongest are protected.

The convergence analysis provides theoretical and empirical justification for the numerical pipeline: Kaniel–Saad bounds show that $m = 10$ Lanczos iterations suffice for spike recovery, and the subspace stability analysis demonstrates robustness to stochastic Hessian estimation. The bulk-walk experiment provides independent validation that class-discriminative information resides exclusively in the spike subspace—9 directions out of 23 million parameters.

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